

Trainings ▾

General Information

WARNING

Never attempt to rotate the engine by pulling or prying on the fan. This practice will result in serious personal injury and damage to the fan. Use only the proper engine barring techniques to manually rotate the engine.

Check the fan for missing balance weights at each regular maintenance interval ¹. Do **not** attempt to repair broken or bent fans, or fans with missing balance weights.

The majority of equipment that has a Cummins® engine has a radiator and a fan. The radiator and fan transfer heat from the cooling water to the atmosphere. The fan selection process **must** conclude that the fan, the fan mounting arrangement, and the fan drive system are designed and matched for compatibility.

Upon request, Cummins Application Engineering Department will assist in determining the proper selection. Contact the Cummins Application Engineering Department for prior approval for any fan changes, other than the direct replacement of a fan with precisely the same Cummins® part number.

Examples that require approval are:

- Using an approved fan from one engine model on a different engine model.
- Using an approved fan on an engine with a different fan mounting arrangement.
- Using an approved fan on an engine with a different fan drive arrangement.
- Converting an engine from one market model to another. An example is the conversion of a G-drive engine to a power unit application.
- Converting an engine model to a different engine model. An example is converting a N-855 to a NT-855 model.

These examples are **not** inclusive. **Always** contact Application Engineering for assistance.

There are times when an existing fan can yield **only** marginal cooling capability when being considered for a new application.

WARNING

Never repitch (bend) the blades to obtain additional air delivery. Bending the blades or spider creates stress in the construction material of the fan. Repitching (bending) will cause fan failure and can cause serious personal injury. The proper diameter fan must be selected. Never modify an existing fan.

Application Engineering will provide assistance in the selection of a fan with the correct pitch and diameter for proper cooling.

¹ Recommended maintenance interval is to every 250 hours or 6 months of operation.

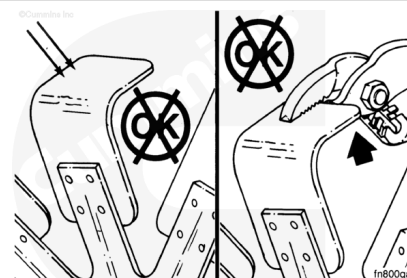
Initial Check

Check the fan for cracked, bent, or broken blades.



⚠ WARNING ⚠

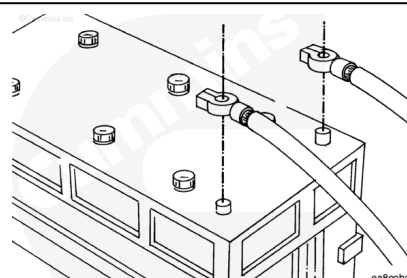
Do not straighten a bent fan blade, or continue to use a damaged fan. A bent or damaged fan blade can fail during operation and cause personal injury or property damage.



Remove

⚠ WARNING ⚠

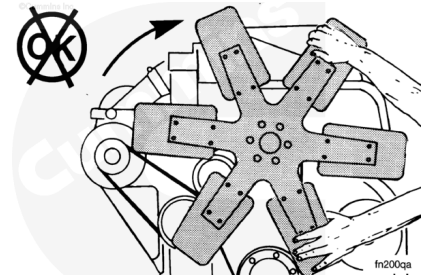
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



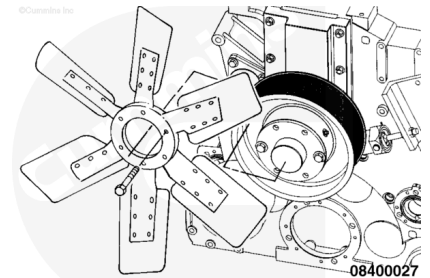
Disconnect the batteries or air supply to the air starter to prevent accidental engine starting. Refer to the OEM troubleshooting and repair manual.

⚠ CAUTION ⚠

Do not pry on the fan. The fan blades can be damaged.

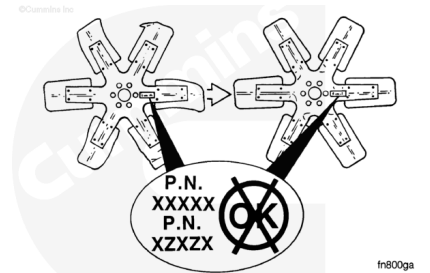


Remove the fan and spacer.



Install

Replace the original equipment fan with a fan of the identical part number. Cummins® **must** approve any fan changes.

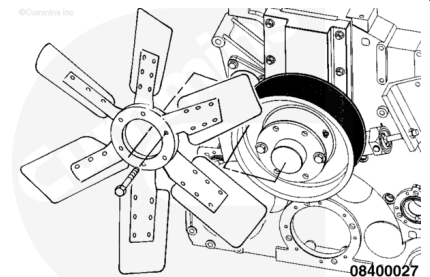


Install the fan and applicable spacer.

Tighten the mounting capscrews.

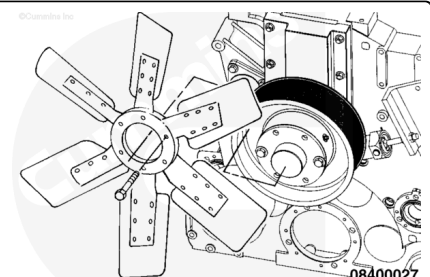
Torque Value: 135 n•m [100 ft-lb]

A minimum of 0.75 mm [3/4-inch] of thread **must** be engaged in the fan hub.



⚠ WARNING ⚠

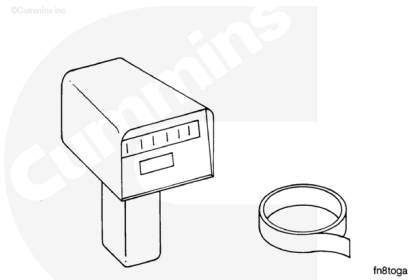
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries or air starter supply line. Refer to OEM troubleshooting and repair manual.

Rotation Check

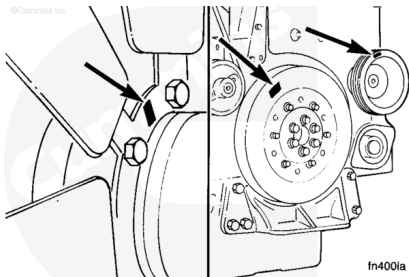
Use an optical tachometer, Part Number 3377462 or equivalent, or a strobe light to measure the fan rpm.



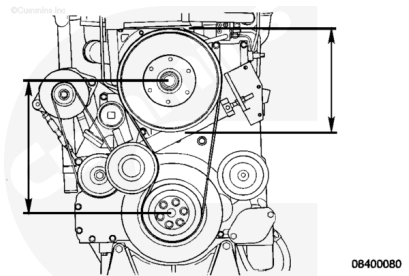
Mark a spot on the fan so the fan rpm can be measured.

Mark another spot on the vibration damper or accessory drive pulley to measure engine rpm.

Note : When using Cummins® optical tachometer, Part Number 3377462, use a piece of reflective tape, Part Number 3377464, to mark the spot.



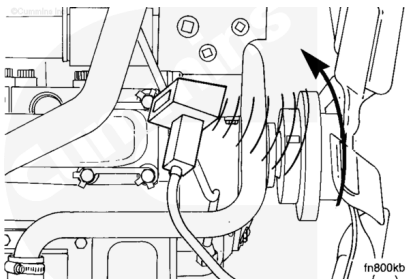
Measure the diameter of the fan belt pulley to determine the drive ratio.



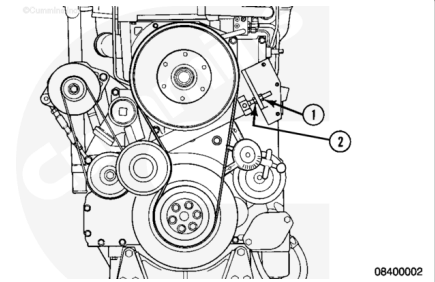
Belt Driven Fan	
Fan Pulley Diameter	Fan rpm
239 mm [9.4 in]	.85 x engine rpm
289 mm [11.4 in]	.70 x engine rpm
338 mm [13.3 in]	.60 x engine rpm

Note : The fan center distance is the distance between the center of the fan hub and the center of the crankshaft.

Operate the engine. Read and compare the fan rpm to the engine rpm. Compare the fan rpm to the specification.



If the fan rpm is not correct, check the belt tension. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>).



Last Modified: 30-Jun-2006
